

TITLE

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Abstract

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1 Introduction

2 Literature Review

2.1 Anti-Money Laundering & Fraud Detection

2.2 Graph Neural Networks

2.2.1 Heterogeneous Graph Transformer (HGT)

The Heterogeneous Graph Transformer (HGT) introduced by Hu et al. [2020] extends the transformer architecture to heterogeneous graphs containing multiple node and edge types. Unlike homogeneous GNNs that treat all nodes uniformly, HGT maintains type-specific parameters for attention computation, allowing the model to learn distinct relationship patterns across different edge types. The architecture uses meta-relation triplets (source type, edge type, target type) to parameterize attention heads, enabling the model to capture semantic differences between relationship types. HGT demonstrated strong performance on large-scale heterogeneous graphs with millions of nodes and billions of edges, making it particularly suitable for financial transaction networks where accounts and transactions form natural heterogeneous structures.

2.2.2 Heterogeneous Graph Masked Autoencoders (HGMAE)

Tian et al. [2023] propose HGMAE, a self-supervised pre-training framework for heterogeneous graphs based on masked autoencoding. The approach masks portions of the graph structure and node features, then trains the model to reconstruct the masked elements. This pre-training strategy enables the model to learn rich node embeddings that capture both structural patterns and semantic relationships without requiring labeled data. The learned embeddings can subsequently be used for downstream classification tasks with a simple classifier. For transaction networks with severe class imbalance, HGMAE offers the advantage of leveraging the full unlabeled graph structure during pre-training, potentially learning patterns that supervised approaches might miss due to the scarcity of positive examples.

2.2.3 Heterogeneous Message Passing Neural Network (HMPNN)

Johannessen and Jullum [2023] specifically address money laundering detection using heterogeneous graph neural networks. Their HMPNN framework extends traditional message passing to heterogeneous graphs by defining separate message functions for each meta-step (source type, edge type, target type). The model operates directly on edge features such as transaction amounts and timestamps, making it well-suited for financial data where transaction attributes carry critical information. Two aggregation variants are proposed: HMPNN-sum, which sums messages across meta-paths, and HMPNN-ct, which concatenates them to preserve distinct relationship signals. Experiments on synthetic AML datasets demonstrate that HMPNN-ct outperforms both homogeneous GNN baselines and traditional machine learning approaches, highlighting the importance of heterogeneous graph modeling for detecting complex laundering patterns.

2.3 Explainable AI in Graph Settings

References

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